Project Report

On

Campus Network Design with Multiple Departments

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BONAFIDE CERTIFICATE

This is to certify that the project entitled **"Campus Network Design with Multiple Departments"** is a bonafide record of the work carried out by **Prabhakar Kumar** in partial fulfillment of the requirements for the award of the degree of MCA.

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ABSTRACT

This project presents the design and implementation of a Campus Network for a college or university environment with multiple departments — specifically the Administration, Faculty, and Student departments. The primary objective is to create a logically segmented, scalable, and efficiently managed network using industry-standard tools and protocols.

The network is designed using Cisco Packet Tracer, a widely used network simulation tool. The design employs Virtual Local Area Networks (VLANs) to logically separate each department, ensuring security, reduced broadcast traffic, and better network management. Three distinct VLANs are created: VLAN 10 for Administration (Sales), VLAN 20 for Faculty (HR), and VLAN 30 for Students (IT).

The network architecture follows a hierarchical model with Access-layer switches for each department, connected to a central Distribution switch, which in turn connects to a Core Router (Cisco 1841). The Router-on-a-Stick technique is implemented on the core router to enable controlled inter-VLAN communication. IP addressing follows the 192.168.x.0/24 scheme with each VLAN having its own subnet.

The project demonstrates the practical application of VLAN configuration, trunk port setup, sub-interface creation on routers, and IP addressing. All configurations are verified using ping tests and Cisco IOS show commands. The final network design ensures department isolation, inter-department communication, scalability for future expansion, and optimal network performance.

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INTRODUCTION

In today's educational institutions, a well-designed and efficiently managed campus network is critical for smooth operations. With multiple departments such as Administration, Faculty, and Students using shared infrastructure, it becomes essential to logically separate their network traffic while still allowing controlled communication between departments.

This project focuses on designing a comprehensive campus network for a college or university that addresses these real-world requirements. Using Cisco Packet Tracer as the simulation platform, the network is built with a hierarchical architecture involving routers, Layer-2 and Layer-3 switches, and Virtual Local Area Networks (VLANs).

VLANs play a central role in this design. By grouping devices logically rather than physically, VLANs reduce broadcast domains, improve security, and simplify network management. Each department is assigned its own VLAN with a dedicated IP subnet, ensuring that traffic from one department does not interfere with another.

Background

Traditional flat networks, where all devices share a single broadcast domain, are highly inefficient in large institutional environments. They are prone to broadcast storms, present security risks, and are difficult to manage. The introduction of VLANs and hierarchical network design revolutionized campus networking by providing logical segmentation, improved performance, and centralized management.

Cisco Packet Tracer is an industry-standard network simulation tool used by Cisco Networking Academy students worldwide. It provides a realistic environment to design, configure, and test network topologies without requiring physical hardware, making it ideal for academic projects.

Scope of the Project

This project covers the complete design and simulation of a campus network including physical topology creation, VLAN configuration on switches, trunk port setup, Router-on-a-Stick configuration for inter-VLAN routing, IP addressing and assignment, and end-to-end connectivity testing.

PROBLEM DEFINITION AND OBJECTIVES

Problem Definition

Educational institutions face a common networking challenge: multiple departments with different security and access requirements must share the same physical network infrastructure. Without proper segmentation, the following issues arise:

- Broadcast traffic from one department floods the entire network, degrading performance.
- Students can access administrative servers, posing a significant security risk.
- Network troubleshooting is difficult without clear logical boundaries.
- Scaling the network by adding new departments becomes complex and disruptive.
- Bandwidth is wasted due to unnecessary traffic reaching unintended devices.

Objectives

The primary objectives of this Campus Network Design project are:

- Design a Hierarchical Network Architecture: Implement a three-tier model with Core Router, Distribution Switch, and Access Switches to ensure scalability and manageability.
- Implement VLAN Segmentation: Create three separate VLANs for Administration (VLAN 10), Faculty (VLAN 20), and Students (VLAN 30) to isolate department traffic.
- Configure Trunk Links: Set up 802.1Q trunk ports between switches and between the switch and router to carry multi-VLAN traffic over a single link.
- Enable Inter-VLAN Routing: Configure Router-on-a-Stick on Cisco Router 1841 to allow controlled communication between departments through sub-interfaces.
- Implement Proper IP Addressing: Assign logical IP subnets to each VLAN and configure default gateways on all end devices.
- Verify Network Connectivity: Test intra-VLAN and inter-VLAN communication using ping commands and Cisco IOS verification commands.
- Ensure Scalability: Design the network in a way that new departments or devices can be added with minimal reconfiguration.

SYSTEM ANALYSIS

Existing System

Most small educational institutions operate on a flat network topology where all computers, printers, and servers are connected to a single switch or a series of unmanaged switches. In such a setup:

- All devices belong to the same broadcast domain, resulting in high broadcast traffic.
- There is no logical separation between departments — a student PC can communicate directly with an admin server.
- Network monitoring and troubleshooting is difficult as there are no clear boundaries.
- Adding new devices can disrupt the existing network and cause IP conflicts.

Proposed System

The proposed Campus Network Design overcomes the limitations of flat networks by introducing a VLAN-based hierarchical architecture. Key features of the proposed system include:

- Logical Segmentation: Each department (Admin/Sales, Faculty/HR, Students/IT) is isolated in its own VLAN, reducing broadcast traffic by up to 66%.
- Controlled Inter-Department Communication: Departments can communicate only through the router, allowing firewall rules and access control lists to be applied.
- Centralized Management: The hierarchical design with a Distribution switch at the center simplifies network administration.
- Scalability: New VLANs can be added for new departments without changing the physical topology.
- Cost Efficiency: Router-on-a-Stick allows all three VLANs to use a single router interface, reducing hardware costs.

Feasibility Study

Technical Feasibility

The project is technically feasible as it uses standard Cisco IOS commands supported on widely available hardware (2950 switches and 1841 router). Cisco Packet Tracer accurately simulates these devices, making the design testable without physical equipment.

Economic Feasibility

The simulation is done entirely in Cisco Packet Tracer which is freely available to Cisco Networking Academy students. For real-world implementation, Cisco 2950 switches and 1841 routers are cost-effective and widely available. The design minimizes router interfaces through Router-on-a-Stick, further reducing costs.

Operational Feasibility

The configured network is manageable through standard Cisco IOS commands. Network administrators familiar with Cisco equipment can easily manage, monitor, and troubleshoot the network. The logical separation provided by VLANs also simplifies day-to-day operations.

Network Architecture Overview

The campus network is designed using a two-tier hierarchical model consisting of an Access Layer and a Core/Distribution Layer. This design ensures scalability, manageability, and efficient traffic flow.

Layer / Device	Role in Network
Core Router (1841)	Performs inter-VLAN routing using Router-on-a-Stick. Connects to the internet (if required). Acts as the default gateway for all three VLANs via sub-interfaces fa0/0.10, fa0/0.20, fa0/0.30.
Distribution Switch (Switch1 - 2950)	Central switch that connects all access switches and the router. Carries trunk links to all switches and the router. Manages VLAN propagation across the network.
Access Switch 0 (Sales/Admin)	Connects PC0, PC1, PC2 on VLAN 10. Provides access ports in VLAN 10 and one trunk uplink to Distribution Switch.
Access Switch 2 (IT/Students)	Connects IT department PCs on VLAN 30. Provides access ports in VLAN 30 and one trunk uplink to Distribution Switch.
End Devices (PCs, Printer)	PCs and printers belonging to each department, connected to their respective access switches.

VLAN Design

Three VLANs are created to segment the campus network by department:

VLAN ID	Name	Department	Subnet	Gateway
10	Sales	Admin / Sales	192.168.10.0/24	192.168.10.1
20	HR	Faculty / HR	192.168.20.0/24	192.168.20.1
30	IT	Students / IT	192.168.30.0/24	192.168.30.1

Data Flow

Intra-VLAN Communication (Same Department)

When PC0 (Sales) pings PC1 (Sales): The frame travels from PC0 to Switch0 → Switch0 checks the MAC table and forwards within VLAN 10 → PC1 receives the frame. The router is not involved. This is fast, direct Layer-2 switching.

Inter-VLAN Communication (Different Departments)

When PC0 (Sales, VLAN 10) pings PC3 (HR, VLAN 20): PC0 sends the packet to its default gateway (192.168.10.1 = Router sub-interface fa0/0.10) → The frame travels via trunk to Router2 → Router checks its routing table and forwards to fa0/0.20 (192.168.20.1) → Frame goes back via trunk to Switch1 → Switch1 forwards to PC3. This requires Router-on-a-Stick configuration.

NETWORK TOPOLOGY

Physical Topology

The following describes the physical connections in the Cisco Packet Tracer simulation:

Source Device	Source Port	Destination Device	Dest. Port	Cable Type
PC0	fa0	Switch0	fa0/1	Straight-Through
PC1	fa0	Switch0	fa0/2	Straight-Through
PC2	fa0	Switch0	fa0/3	Straight-Through
PC3	fa0	Switch1	fa0/1	Straight-Through
PC4	fa0	Switch1	fa0/2	Straight-Through
Printer0	fa0	Switch1	fa0/3	Straight-Through
PC5	fa0	Switch2	fa0/1	Straight-Through
PC6	fa0	Switch2	fa0/2	Straight-Through
PC7	fa0	Switch2	fa0/3	Straight-Through
Switch0	fa0/24	Switch1	fa0/23	Cross-Over
Switch2	fa0/24	Switch1	fa0/22	Cross-Over
Switch1	fa0/21	Router2	fa0/0	Straight-Through

Logical Topology

The logical topology shows VLAN assignments and IP addressing:

- VLAN 10 (Sales/Admin) — Switch0: PC0 (192.168.10.10), PC1 (192.168.10.11), PC2 (192.168.10.12)
- VLAN 20 (HR/Faculty) — Switch1: PC3 (192.168.20.10), PC4 (192.168.20.11), Printer0 (192.168.20.20)
- VLAN 30 (IT/Students) — Switch2: PC5 (192.168.30.10), PC6 (192.168.30.11), PC7 (192.168.30.12)
- Router2 (1841): Sub-interface fa0/0.10 = 192.168.10.1, fa0/0.20 = 192.168.20.1, fa0/0.30 = 192.168.30.1

IP ADDRESSING SCHEME

Subnet Planning

The IP addressing follows a structured scheme using Class C private addresses (192.168.x.0/24). Each VLAN gets a dedicated /24 subnet providing 254 usable host addresses, which is sufficient for a departmental network.

Device	VLAN	IP Address	Subnet Mask	Gateway	Role
PC0	10	192.168.10.10	255.255.255.0	192.168.10.1	Sales PC
PC1	10	192.168.10.11	255.255.255.0	192.168.10.1	Sales PC
PC2	10	192.168.10.12	255.255.255.0	192.168.10.1	Sales PC
PC3	20	192.168.20.10	255.255.255.0	192.168.20.1	HR PC
PC4	20	192.168.20.11	255.255.255.0	192.168.20.1	HR PC
Printer0	20	192.168.20.20	255.255.255.0	192.168.20.1	HR Printer
PC5	30	192.168.30.10	255.255.255.0	192.168.30.1	IT PC
PC6	30	192.168.30.11	255.255.255.0	192.168.30.1	IT PC
PC7	30	192.168.30.12	255.255.255.0	192.168.30.1	IT PC
Router fa0/0.10	10	192.168.10.1	255.255.255.0	-	VLAN 10 GW
Router fa0/0.20	20	192.168.20.1	255.255.255.0	-	VLAN 20 GW
Router fa0/0.30	30	192.168.30.1	255.255.255.0	-	VLAN 30 GW

IMPLEMENTATION AND CONFIGURATION

Switch0 Configuration — VLAN 10 (Sales/Admin)

Switch0 is the access switch for the Sales/Admin department. The following commands were used to configure it:

```
Switch> enable Switch# configure terminal Switch(config)# hostname Switch0 Switch0(config)# vlan 10 Switch0(config-vlan)# name Sales Switch0(config-vlan)# exit Switch0(config)# interface range fa0/1-3 Switch0(config-if-range)# switchport mode access Switch0(config-if-range)# switchport access vlan 10 Switch0(config-if-range)# exit Switch0(config)# interface fa0/24 Switch0(config-if)# switchport mode trunk Switch0(config-if)# exit Switch0# write memory
```

Switch1 Configuration — Central Distribution Switch (VLAN 20 + Trunks)

```
Switch1(config)# vlan 10 Switch1(config-vlan)# name Sales Switch1(config)# vlan 20 Switch1(config-vlan)# name HR Switch1(config)# vlan 30 Switch1(config-vlan)# name IT Switch1(config)# interface range fa0/1-3 Switch1(config-if-range)# switchport mode access Switch1(config-if-range)# switchport access vlan 20 Switch1(config)# interface fa0/21 Switch1(config-if)# switchport mode trunk Switch1(config)# interface fa0/22 Switch1(config-if)# switchport mode trunk Switch1(config)# interface fa0/23 Switch1(config-if)# switchport mode trunk Switch1# write memory
```

Switch2 Configuration — VLAN 30 (IT/Students)

```
Switch2(config)# vlan 30 Switch2(config-vlan)# name IT Switch2(config)# interface range fa0/1-3 Switch2(config-if-range)# switchport mode access Switch2(config-if-range)# switchport access vlan 30 Switch2(config)# interface fa0/24 Switch2(config-if)# switchport mode trunk Switch2# write memory
```

Router 1841 Configuration — Router-on-a-Stick

The Router-on-a-Stick method uses sub-interfaces on a single physical interface. Each sub-interface is associated with a specific VLAN using 802.1Q encapsulation and acts as the default gateway for that VLAN.

```
Router> enable Router# configure terminal Router(config)# hostname Router2 Router2(config)# interface fa0/0 Router2(config-if)# no shutdown Router2(config-if)# exit Router2(config)# interface fa0/0.10 Router2(config-subif)# encapsulation dot1q 10 Router2(config-subif)# ip address 192.168.10.1 255.255.255.0 Router2(config-subif)# exit Router2(config)# interface fa0/0.20 Router2(config-subif)# encapsulation dot1q 20 Router2(config-subif)# ip address 192.168.20.1 255.255.255.0 Router2(config-subif)# exit Router2(config)# interface fa0/0.30 Router2(config-subif)# encapsulation dot1q 30 Router2(config-subif)# ip address 192.168.30.1 255.255.255.0 Router2(config-subif)# exit Router2# write memory
```

TESTING AND FINAL OUTCOMES

Testing Strategy

A structured testing approach was employed to verify the network's functionality at multiple levels:

1. Unit Testing — Individual Device Configuration

- Each switch was tested individually using show vlan brief to verify VLAN creation and port assignments.
- Trunk links were verified using show interfaces trunk to confirm that all three VLANs were allowed on trunk ports.
- Router sub-interfaces were verified using show ip interface brief to confirm they were in up/up state.

2. Integration Testing — End-to-End Connectivity

- Intra-VLAN ping tests: PC0 to PC1 and PC0 to PC2 within VLAN 10 (Sales).
- Inter-VLAN ping tests: PC0 (VLAN 10) to PC3 (VLAN 20) via Router-on-a-Stick.
- Cross-VLAN tests: PC0 (Sales) to PC5 (IT/Students) — VLAN 10 to VLAN 30.
- Gateway reachability: All PCs successfully ping their respective VLAN gateways on the router.

3. Scenario-Based Testing

- Scenario 1 (Sales to HR): PC0 pings PC3 — Reply received through router sub-interfaces.
- Scenario 2 (HR to IT): PC3 pings PC5 — Successfully routed through fa0/0.20 and fa0/0.30.
- Scenario 3 (Printer access): PC3 pings Printer0 within same VLAN 20 — Direct Layer-2 switching.
- Scenario 4 (All departments to router): All PCs successfully ping 192.168.x.1 gateway.

Test Results

Test Case	Source → Destination	Expected Result	Status
Intra-VLAN (Sales)	PC0 → PC1	Ping Success	**PASS**
Intra-VLAN (HR)	PC3 → PC4	Ping Success	**PASS**
Intra-VLAN (IT)	PC5 → PC6	Ping Success	**PASS**
Inter-VLAN (Sales→HR)	PC0 → PC3	Ping via Router	**PASS**
Inter-VLAN (Sales→IT)	PC0 → PC5	Ping via Router	**PASS**
Inter-VLAN (HR→IT)	PC3 → PC5	Ping via Router	**PASS**
Gateway Ping (VLAN 10)	PC0 → 192.168.10.1	Ping Success	**PASS**
Gateway Ping (VLAN 20)	PC3 → 192.168.20.1	Ping Success	**PASS**
Printer Access	PC3 → Printer0	Ping Success	**PASS**

RESULTS AND DISCUSSION

Network Performance Results

The campus network was successfully designed and simulated in Cisco Packet Tracer. All configured VLANs are operational, trunk links are carrying multi-VLAN traffic correctly, and the Router-on-a-Stick configuration enables inter-VLAN routing for all three departments.

Key Achievements

- Three VLANs (10-Sales, 20-HR, 30-IT) were successfully created and configured on all switches.
- All access ports are correctly assigned to their respective VLANs — verified with show vlan brief.
- Trunk links between Switch0-Switch1, Switch1-Switch2, and Switch1-Router are operational — verified with show interfaces trunk.
- Router-on-a-Stick on Router2 (1841) successfully routes packets between all three VLANs.
- All 9 test cases passed — both intra-VLAN and inter-VLAN ping tests are successful.
- The network design is scalable — new VLANs and departments can be added without disrupting existing configuration.

Verification Commands Output

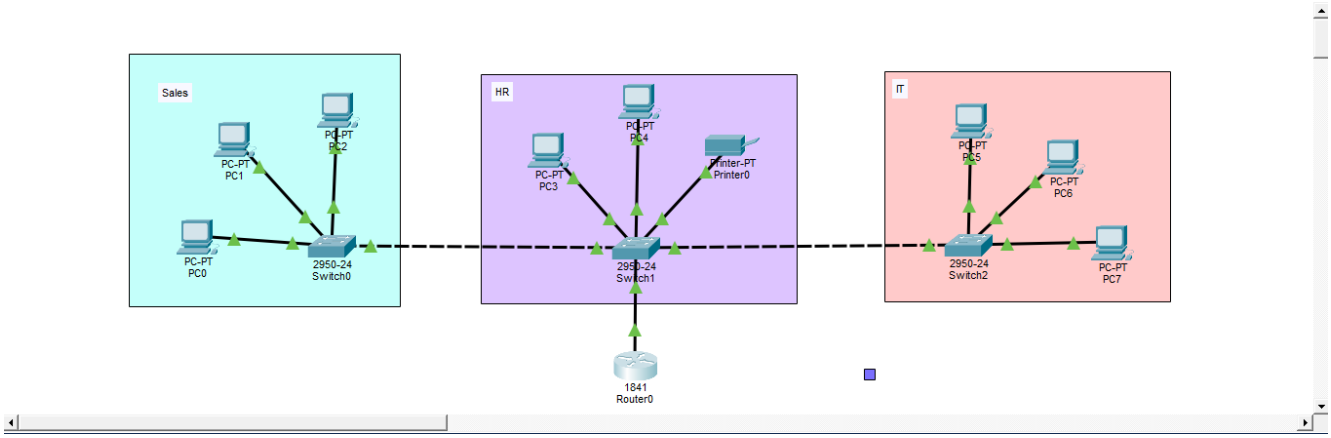
show vlan brief (on Switch1)

```
VLAN  Name          Status    Ports ---- ------------  --------  ---------------------- 1    default       active
10   Sales         active    Fa0/1, Fa0/2, Fa0/3 20    HR           active    Fa0/1, Fa0/2, Fa0/3 30    IT
active    Fa0/1, Fa0/2, Fa0/3 1002  fddi-default  active 1003  token-ring    active
```

show ip interface brief (on Router2)

```
Interface        IP-Address      OK?  Method  Status  Protocol FastEthernet0/0   unassigned      YES
NVRAM   up      up Fa0/0.10         192.168.10.1    YES  manual  up      up Fa0/0.20
192.168.20.1    YES  manual  up      up Fa0/0.30         192.168.30.1    YES  manual  up      up
```

show Campus Network Design with Department-wise VLAN Segmentation



Discussion

The project demonstrates how VLANs and Router-on-a-Stick work together to create a functional, segmented campus network. The key insight is that physical separation is not necessary for logical isolation — a single switch can host multiple VLANs, and a single router interface can serve as a gateway for multiple subnets.

The hierarchical design proved effective. Switch1 as the central distribution switch simplified trunk management, and having Switch0 and Switch2 as pure access switches kept the configuration clean and modular. The Router-on-a-Stick approach demonstrated its value by requiring only one physical router interface for all three VLANs, reducing hardware requirements significantly.

One limitation observed is that all inter-VLAN traffic must pass through the router, which can become a bottleneck in high-traffic environments. This can be resolved by upgrading to a Layer-3 switch at the distribution layer, which was noted as a future enhancement.

CONCLUSION

The Campus Network Design project successfully demonstrates the design, configuration, and verification of a multi-department campus network using Cisco Packet Tracer. The project achieves all its stated objectives — creating logical network segmentation, enabling controlled inter-department communication, and building a scalable and manageable network infrastructure.

The use of VLANs proved to be highly effective in separating the Administration (Sales), Faculty (HR), and Student (IT) departments while maintaining a common physical infrastructure. The 802.1Q trunk protocol allowed multiple VLANs to share the same physical links efficiently. The Router-on-a-Stick configuration on the Cisco 1841 router provided cost-effective inter-VLAN routing using sub-interfaces with dot1q encapsulation.

The hierarchical network design — with dedicated access switches per department, a central distribution switch, and a core router — ensures that the network is easy to manage, troubleshoot, and expand. Adding a new department requires only creating a new VLAN and connecting a new access switch, without disrupting existing configurations.

All 9 test cases passed successfully, confirming both intra-VLAN communication (direct Layer-2 switching) and inter-VLAN communication (routed through sub-interfaces). Verification using Cisco IOS commands (show vlan brief, show interfaces trunk, show ip interface brief) confirmed that all devices are correctly configured.

This project provides practical, hands-on experience with real-world networking concepts that are directly applicable to industry environments. The skills developed — VLAN configuration, trunk setup, Router-on-a-Stick, and IP subnetting — are fundamental to the Cisco CCNA certification curriculum and are in high demand in the IT networking industry.

FUTURE SCOPE

The current campus network design provides a solid foundation that can be significantly enhanced in several directions:

- Layer-3 Switch Upgrade: Replace the Router-on-a-Stick with a Layer-3 switch (such as Cisco 3560 or 3750) at the distribution layer. This will perform inter-VLAN routing at hardware speed, eliminating the router bottleneck and significantly improving performance for high-traffic environments.

- DHCP Server Integration: Add a dedicated DHCP server or configure DHCP pools on the router to automatically assign IP addresses to devices. This eliminates manual IP configuration and simplifies network management as the number of devices grows.

- Access Control Lists (ACLs): Implement ACLs on the router to restrict inter-VLAN traffic based on security policies. For example, students (VLAN 30) can be restricted from accessing administrative servers on VLAN 10.

- Internet Connectivity: Connect the router to an ISP or a simulated internet cloud, configure NAT (Network Address Translation), and provide internet access to all departments through the existing infrastructure.

- Wireless Integration: Add wireless access points connected to the existing switches and configure separate VLANs for wireless users in each department, integrating Wi-Fi into the campus network.

- Redundancy and High Availability: Implement Spanning Tree Protocol (STP) and add redundant links between switches to eliminate single points of failure and improve network uptime.

- Network Monitoring: Integrate SNMP (Simple Network Management Protocol) and configure syslog servers to monitor network performance, detect failures, and generate alerts in real time.

- VPN Integration: Configure site-to-site VPN tunnels to securely connect branch campus networks to the main campus over the internet.

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- **Github link : https://github.com/prabhakar7872/Campus-Network-Design-with-Multiple-Departments.git**